

LEDGER · 2026-09-04 · MATERIALS SCIENCE

Formaldehyde-Free Tannin-Furanic Rigid Foam via Phytic Acid Synergistic Crosslinking

REJECTED

This concept did **not** clear the framework.
The reasoning is published in full below.
What the assessment found

Incumbent benchmark	Rigid Polyurethane Foam (RPUF)
Entry application	factory-manufactured, fire-rated core insulation for structural sandwich panels
Measured advantage	>2.8x improvement in Limiting Oxygen Index (LOI >70% vs RPUF ~24.8%)
Time to first revenue	24 months
Gross margin at parity	11.9%
Startup CapEx	\$350,000
Payback from first sale	14.3 months
Capital productivity	10.05x

Why it failed

- Voided by: Mismatched comparison, Capital and time to revenue, Economics undisclosed, Form-factor / handling regression, Absence search missing
- Evidence rests on non-peer-reviewed sources
- Margin fails the operating-cost check
- payback_over
- Absence asserted, search unreported

Assessment

The concept fails multiple mandatory venture gates, as explicitly conceded in the report: gross margins are ~12% (below 70%), CapEx is \$350,000 (exceeds \$250,000), and time to revenue is 24 months (exceeds 18 months). Furthermore, it fails form factor parity due to severe handling regressions versus liquid PU, and lacks a structured Market Absence Ledger with numeric counts.

Also flagged

- Duplicate references inflating the evidence base
- Price parity asserted, not sourced

A rejection is not a claim that the science is wrong. It means the venture case did not survive: the comparator was not what the buyer actually buys, the comparison was not like-for-like, the advantage fell short of a step change, or the capital and time to first revenue were prohibitive.

Assessed 2026-09-04 against seven gates plus the voiding sub-tests. [How the method works](#) · [Browse all concepts](#)

published scientific literature and have not been commercially validated; several involve chemical, biological or regulated processes that require appropriate expertise, facilities, permits and safety controls. Independently verify every figure, citation and regulatory requirement before committing capital.